Software Requirements Specification

for

Fill My Fridge App

Version 1.0 final

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December 12th, 2024

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Revision History

Name	Date	Reason For Changes	Version
Billy, Ana, David, Yuliya	9/30/2024	First draft of the SRS	1.0 draft
David	10/7/2024	Added glossary and Stimulus/Response sequences	1.0 draft II
Ana, Billy, David, Yuliya	10/21/2024	Added functional requirements Added GUI prototypes Added ERD, Crud Matrix, and DFD Diagrams	1.0 draft III
Ana, Billy, David, Yuliya	10/28/2024	Added quality attributes Added internationalization and localization requirements Updated constraints Updated use cases	1.0 draft IV
Ana, Billy, David, Yuliya	11/04/2024	Updated based on defect report.	1.0 approved
Ana	12/12/2024	Added sections 4.3 and 5.2 Updated Table of Contents to have all sections defined Updated header to reflect our project name	1.0 final

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1. Introduction

1.1 Purpose

This SRS describes the Functional and Nonfunctional Requirements for software release 1.0 of the Fill My Fridge application. This document is intended to be used by the members of the project team who will implement and verify the correct functioning of the system. Unless otherwise noted, all requirements specified here are committed for release 1.0.

1.2 Document Conventions

This SRS has no unordinary typographical conventions or standards used.

1.3 Project Scope

The Fill My Fridge App will contain features that allow users to add Recipes and set Nutrition Goals, then using these features to generate a Meal Plan tailored to their needs. They will even be able to create a Grocery List from these meal plans they have generated. A more thorough look into these features along with their implementation is detailed in the Fill My Fridge Vision and Scope Document [1].

1.4 References

[1] Team 4, *Fill My Fridge Vision and Scope Document, Version 1.0 approved,* https://github.com/yorker78/fill-my-fridge/blob/main/docs/Vision%20and%20Scope.pdf
[2] Belinski, E. (n.d.). Android API Levels. API Levels | Android versions, SDK/API levels, version codes, codenames, and cumulative usage. https://apilevels.com/
[3] Belinski, E. (n.d.-b). IOS version usage - ios ref. iOS version usage -. https://iosref.com/ios-usage
[4] Google. (n.d.). Samsung Food: Meal Planning - apps on Google Play. Google. https://play.google.com/store/apps/details?id=com.foodient.whisk
[5] Wells, J. (2020, November 16). How 6 chains expanded pickup during the pandemic. Grocery Dive. https://www.grocerydive.com/news/how-6-chains-expanded-pickup-during-the-pandemic/589054/
[6] Coding conventions: Kotlin. Kotlin Help. (n.d.). https://kotlinlang.org/docs/coding-conventions.html

2. Overall Description

2.1 Product Perspective

The Fill My Fridge App is a new app to provide an all-in-one user experience for nutrition tracking, meal planning, and grocery shopping. The Context Diagram in Figure 1 shows the External Systems the app will interact with for release 1.0. Future releases will include a Premium Subscription service connecting to Fill My Fridge servers to allow for Data Syncing and sharing across devices.

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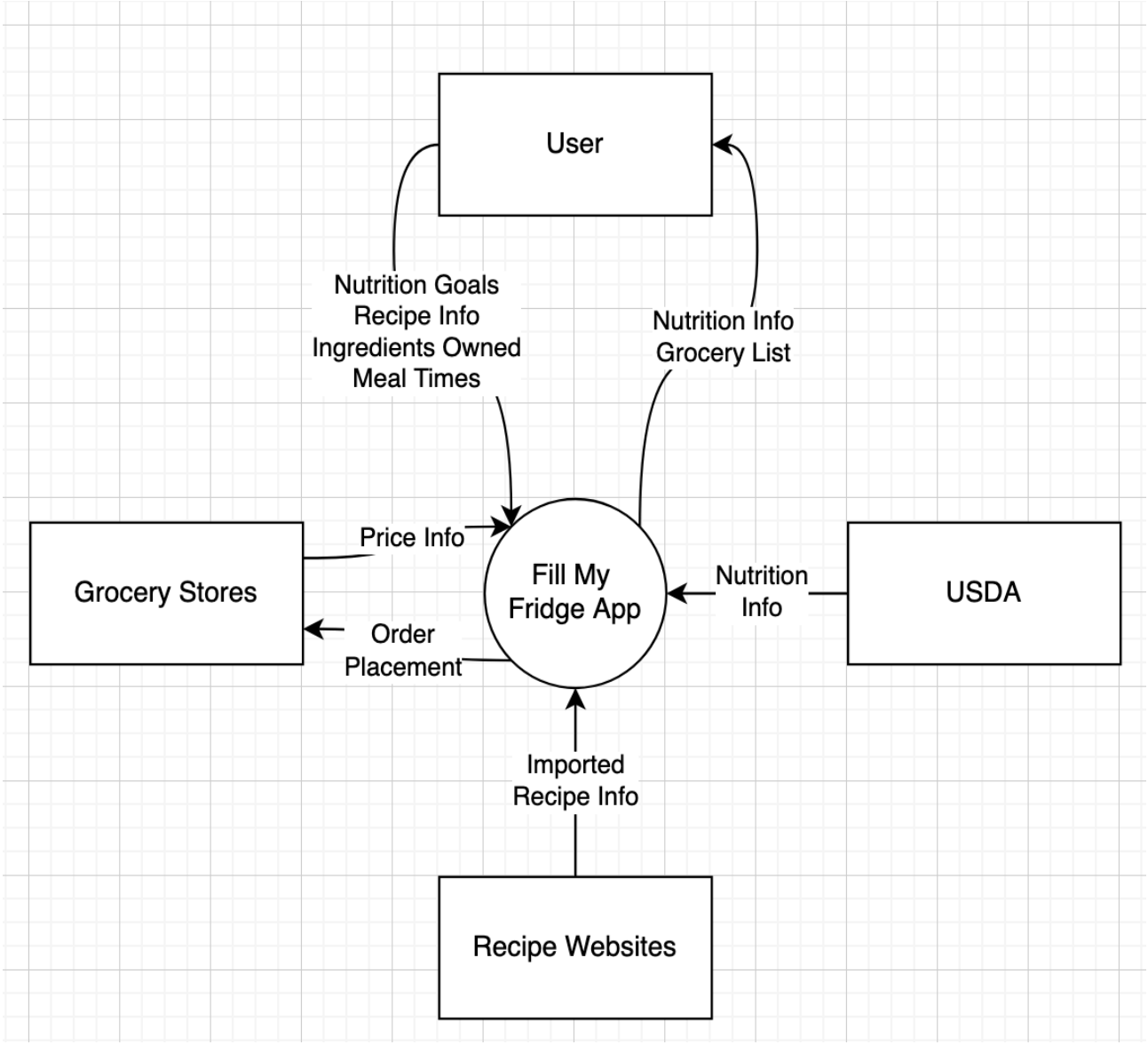


Figure 1. Context Diagram for release 1.0 of the FillMyFridge App

2.2 User Classes and Characteristics

Main Cook in Family	This user is looking for a way to cut down on food and time waisted cooking and shopping. They want to have an easier time creating a plan for meals to eat throughout the week and immediately know what they need to get at the store. This user is more concerned with food waist and time management, they are not as concerned with their nutritional tracking.
Athlete	This user uses Fill My Fridge mostly for the nutritional tracking feature but also uses the mail planning to track their intake. They will end up using the grocery list feature but will not be that dedicated to maintaining the pantry.
Professional chef	This user is looking mostly for a way to cut down on food waist and looking for a way to check the nutritional value of the meals they plan for their restaurant/clients. They are trying to ensure the meals/courses the prepare are meeting the needs of their clients and that they don't over shop to ensure less food waist and not overspending on expensive ingredients.

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2.3 Operating Environment

OE-1: The app shall operate correctly on Android Versions 8 and newer and iOS versions 15 and newer to capture 95%[2][3] of the two most popular mobile application markets.
OE-2: The app shall use the modern Native Development techniques, Kotlin for Android and swift for iOS and be available on the respective "App Stores" for each mobile operating system.
OE-3: The app shall store all user's data locally on device.
OE-4: The app shall store all user created data on the cloud while all Fill My Fridge accompanying data will still be stored locally.

2.4 Design and Implementation Constraints

CO-1: The app shall conform to all policies established by Google and Apple to maintain in good standing with their marketplaces.
CO-2: The Android App Kotlin codebase shall conform to the Coding Conventions as defined by the Kotlin language developers[6]
CO-3: The iOS App Swift codebase shall conform to the coding conventions as defined by Apple
CO-4: The app shall conform to all polices for API consumers established by Kroger, Walmart, Amazon (Whole Foods), and Target, as well as any other grocery store interfaces added after Release 1.0.

2.5 Assumptions and Dependencies

AS-1: The user has or is willing to sign up for accounts associated with the grocery stores they wish to connect to the app to use the price comparison and online ordering features
AS-2: The premium users will have some connection to internet so their data can be backed up.
DE-1: The operation of Fill My Fridge depends on the USDA Nutritional Database having the ingredients found in grocery stores in the United States.
DE-2: The operation of Fill My Fridge depends on Kroger, Walmart, Amazon (Whole Foods), and Target APIs to provide accurate pricing information.

3. System Features

3.1 Recipe Collecting

3.1.1 Description

User can collect, import and add their own recipes. It is possible to search through the added recipes or delete them (Priority: high).

UC ID and Name:	UC-2: Recipe Collecting		
Created By:	Yuliya Kazadoi	Date Created:	09/29/2024
Primary Actor:	User	Secondary	Internal system (nutrition

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		Actors:	tracker)
Trigger:	User indicates desire to add a recipe manually.		
Description:	User wants to manually add a recipe to the app as a digital copy to be saved to their device. User also wants nutrition information about recipe.		
Preconditions:	UC-2-PRE-1: User has a recipe in mind (not creating on the fly). UC-2-PRE-2: Nutrition info has been gathered from USDA database and can be applied. UC-2-PRE-3: User has space on device.		
Postconditions:	POST-1: User has a recipe saved to device. POST-2: Recipe has nutrition info along with instructions. POST-3: Recipes are searchable.		
Normal Flow:	**1.0 Adding New Recipe** 1. User indicates they want to add a new recipe. 2. System prompts user to enter the recipe data. 3. User enters the name of the recipe. 4. System displays the new name 5. User indicates they want to add ingredient. 6. System prompts User to enter the ingredient name. 7. User begins entering the name of the ingredient 8. System provides an autocomplete list of ingredients with names matching the text the User has input 9. User chooses the option they need. (see 1.9.E1) 10. System prompts the User to enter the quantity of the ingredient. 11. User enters the quantity. 12. System prompts the User to confirm adding the ingredient 13. User confirms adding the ingredient. 14. System prompts User to add more ingredients or to add instructions 15. User adds more ingredients (return to step 6) or goes on to adding instructions (continue normal flow). 16. System prompts User for instruction. 17. User enters instruction. 18. System prompts User to add more instructions or to move on to adding recipe metadata. 19. User adds more instructions (return to step 16) or goes on to adding metadata (continue normal flow). 20. System prompts User for recipe serving quantity and estimated cook time 21. User either enters one or both of serving quantity and estimated cook time, or indicates they want to skip entering the data 22. System prompts User for recipe tags 23. User begins entering the name of a recipe tag 24. System provides an autocomplete list of tags with names matching the text the User has input 25. User selects a tag (See 1.25.E2) 26. System prompts user to add more tags or save the recipe 27. User adds more tags (return to step 22) or saves the recipe		

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	(continue normal flow) 28. System saves the recipe to the device. 29. System indicates to the User the recipe has been saved 30. System returns to recipe book view
Alternative Flows:	**1.1 Cancel adding the recipe** 1. Follow normal flow through step 2. 2. User cancels adding a recipe. 3. The use case exits.
Exceptions:	**1.9.E1 The food item is not found in the database** 1. User finishes entering the ingredient name 2. System does not find the ingredient in the food database. 3. System prompts User to create a new ingredient using the name provided 4. User enters nutrition info. 5. System prompts user to save the new ingredient 6. User indicates they want to save the ingredient 7. System saves the ingredient to the database 8. System indicates to the User it has saved the ingredient 9. System returns to the recipe with the new ingredient saved to the recipe, then return to step 10 of normal flow. **1.25.E2 The tag is not found in the database** 1. User finishes entering the tag name 2. System does not find the tag name in the stored tags 3. System prompts User to create a new tag 4. User indicates they want to save the tag 5. System saves the tag 6. System displays the new tag under the list of tags for the recipe
Priority:	High
Frequency of Use:	2-3 Uses per week. New users are expected to have a much greater average usage of this feature. Approximately 7-10 times per week.
Business Rules:	BR-8
Other Information:	1. System will keep a draft saved until the recipe is either created and saved or discarded by user 2. If they create a custom food item as part of recipe creation and cancel that specific recipe creation, the custom food item is still saved
Assumptions:	UC-2-AS-1: Manually adding recipe. UC-2-AS-2: All food items used in recipe can be found in either branded or foundational food USDA databases. UC-2-AS-3: User at minimum fills out a list of ingredients.

3.1.2 Stimulus/Response Sequences

1. The user indicates they want to add a new recipe.
2. The System prompts user with form.
3. The User fills out form.
4. The System stores the completed form.

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5. The System indicates to the User the recipe has been saved
6. The System returns to the recipe book

3.1.3 Functional Requirements

FR-2-1: When the user indicates they wants to add a new recipe, the system shall prompt user with a new recipe creation form.
FR-2-2: If a user wants to add a food item not found in the USDA database to a recipe they are creating, then the system shall prompt the user to create an entry in the database containing the nutrition facts before proceeding with that recipe creation.
FR-2-3: When the user presses cancel on the recipe creation form the system must save the current state of the recipe to the draft location located in the recommend storage location for operating system.
FR-2-4: When the user closes the Fill My Fridge app before save or cancel has been clicked on the recipe creation form the system must save the current state of the recipe to the draft location located in the recommend storage location for operating system.
FR-2-5: When the user clicks save on the recipe creation form, the system shall save the form data to the device in the recommend storage location for operating system.
FR-2-6: If the Fill My Fridge app crashes before save or cancel is pressed the system shall record none of the changes or additions to the recipe creation form.
FR-2-7: The system shall conform its Recipe Data Type as a subset of the open Recipe Schema available on schema.org.

3.2 Pantry Tracking

3.2.1 Description

The app allows users to keep track of their pantry: add the products they have and transfer them from the shopping list (Priority: moderate).

UC ID and Name:	UC-3: Pantry Tracking		
Created By:	Ana Knutson	Date Created:	09/25/2024
Primary Actor:	User	Secondary Actors:	External systems (USDA database) Internal systems (Grocery List)
Trigger:	User indicates that they want to add an additional item to pantry.		
Description:	User wants to track food they have at home that is not accounted for in the app. User wants pantry data to persist.		
Preconditions:	UC-3-PRE-1: User has food item they want to add.		
Postconditions:	UC-3-POST-1: Pantry has item with quantity saved.		
Normal Flow:	**1.0 Add Item to Pantry** 1. User indicates they want to add an additional ingredient to pantry. 2. System prompts User for the name of the ingredient. 3. User begins entering the name of the ingredient. 4. System provides an autocomplete list of ingredients with names matching the text the User has input 5. User selects one of the ingredients. 6. System prompts the User for the quantity of the item.		

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	7. User enters the quantity of the item. 8. System displays preview of the ingredient 9. User indicates they are ready to add the ingredient. 10. System saves the ingredient to the pantry 11. System indicates to the User the ingredient has been saved 12. System returns to the pantry view
Alternative Flows:	**1.1 Cancel Adding Item to Pantry** 4. Follow normal flow through step 2. 5. User cancels adding an item, and use case exits. **1.2 Item is Already in Pantry** 1. Follow normal flow through step 5. 2. System alerts user that this item is already added in the pantry. 3. System asks user if they want to update the quantity of the item. 4. User enters updated quantity of item. 5. User indicated they are ready to update the item. 6. Return to normal flow at step 9.
Exceptions:	**1.1.E1 User Cancels at Another Point During UC-1.0** 1. User wants to cancel after already entering the name of an item or specifying quantity. 2a. User may cancel their action through step 7 of UC1.0. 2b. User cannot cancel once they have completed step 8 of UC1.0. To change what they did, user must go back through the UC1.0 for that item in the pantry to change the quantity. **1.0.E2 User Adds Food Item Not in USDA Database** 1. User wants to add a food item that is not in the USDA database 2. The food item will still be able to be tracked on their device, but it will be tracked as a custom food item
Priority:	Moderate
Frequency of Use:	User will rarely use this feature, approximately 4-5 times per month.
Business Rules:	BR-5, BR-6, BR-14, BR-15
Other Information:	
Assumptions:	UC-3-AS-1: User is adding all food items they bring into house. UC-3-AS-2: User is tracking food intake with the meal plan. UC-3-AS-3: User is manually keeping track of food expiration.

3.2.2 Stimulus/Response Sequences

1. The user indicates they want to add food item to pantry item.
2. The System prompts user with form.
3. The User fills out form.
4. The System stores the completed form.

3.3 Grocery List Tracking

3.3.1 Description

The user can add items to the grocery list manually or automatically generate it from the meal plan, delete the items or mark them when bought (Priority: high)

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UC ID and Name:	UC-1: Grocery List Creation		
Created By:	Ana Knutson	Date Created:	09/25/2024
Primary Actor:	User	Secondary Actors:	External Systems (Grocery store APIs) Internal Systems (Pantry Tracker, Meal Plan, Recipe Book)
Trigger:	User indicates they want to generate a grocery list.		
Description:	User wants a list of items to fulfill their meal plan. User wants this to be an organized list based on templates.		
Preconditions:	UC-1-PRE-1: User has a meal plan created. UC-1-PRE-2: User has recipes saved in recipe book. UC-1-PRE-3: Items already in user's home are tracked in app.		
Postconditions:	UC-1-POST-1: Grocery list is generated. UC-1-POST-2: Grocery list saved on device for later use.		
Normal Flow:	**1.0 Generating a Grocery List** 1. User indicates they want to generate a grocery list. 2. System prompts the user to indicate whether they want to add a meal plan, a recipe to their list, or if they are done 3. User indicates they want to add a meal plan 4. System prompts the User with a list of their created meal plans 5. User indicates the meal plans they want to add to the grocery list 6. System gathers ingredients from recipes of the selected meal plans and adds them to the list. 7. System prompts the User with a list of their created meal plans that have not been saved for the current grocery list 8. User indicates other days they want to add (return to step 6) or indicates they are done selecting days (continue normal flow 9. System displays a preview of the generated grocery list and prompts User to confirm the added data 10. User confirms the list 11. System generates grocery list. 12. System indicates to the user the list has been created 13. System navigates to grocery list view		
Alternative Flows:	**1.1 Select All Created Meal Plans** 1. Follow normal flow through step 2. 2. User selects an option which selects all premade meal plans on their device. 3. Return to step 4 of normal flow. **1.2 Cancel Grocery List Generation** 1. Follow normal flow through step 3. 2. User indicates they want to cancel. 3. Use case is exited and no grocery list is generated. **1.3 Select Recipes instead of Meal Plans for List Generation** 1. Follow normal flow through step 2 2. User indicates they want to add recipes 3. System asks user which recipes they want to add		

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	4. User indicates the recipes they want to add 5. System prompts User to confirm they are done selecting recipes 6. User indicates they are done selecting recipes 7. Return to step 6 of normal flow
Exceptions:	**1.0.E1 User has No Meal Plans** 1. An empty grocery list will be generated, and the user can write in whatever items they desire 2. There will be no ability for the user to indicate they want to select all as there are no meal plans to select from
Priority:	High
Frequency of Use:	Approximately 1 usage per week. 2-3 on the high end.
Business Rules:	BR-7, BR-9
Other Information:	1. New grocery lists will have carryover from previous list in terms of items that weren't indicated as bought 2. Added items to the grocery list will be flagged to change pantry quantity
Assumptions:	UC-1-AS-1: Generating list before going to grocery store. UC-1-AS-2: Manually generating list. UC-1-AS-3: Generating list based off meal plan.

3.3.2 Stimulus/Response Sequences

1. The User indicates they want to the create grocery list.
2. The System pulls meal plan and generates the grocery list.
3. The System displays the grocery list to the User

3.4 Meal Planning

3.4.1 Description

The user can plan meals ahead by adding the recipes he has or generate the plan based on the chosen nutrition goal. He can cancel the plan or edit it. (Priority: moderate)

UC ID and Name:	UC-4 Add a Recipe to Meal Plan		
Created By:	Billy York	Date Created:	09/29/2024
Primary Actor:	User	Secondary Actors:	Internal Systems (Recipe Book, Nutrition Tracker)
Trigger:	User indicates they want to add a recipe to their meal plan.		
Description:	A user navigates to their meal plan and indicates they want to add a recipe to their meal plan on a specific date time so the system can add it to the total nutrition of the date.		
Preconditions:	UC-4-PRE-1. User has set nutrition goals.		
Postconditions:	UC-4-POST-1. Recipe is added to Meal Plan on selected date time. UC-4-POST-2. Nutrition info for selected date is updated.		
Normal Flow:	**1.0 Add a Single Recipe to Meal Plan** 1. User indicates they want to add a recipe to their meal plan. 2. System prompts user for a date time for the recipe to be eaten. 3. User inputs a date time. 4. System prompts the user to select a recipe. 5. User selects a recipe (see 1.1, 1.2).		

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	6. System updates meal plan and projected nutrition for the date time.
Alternative Flows:	**1.1 Add a New Recipe to Meal Plan** 1. User selects to add a new recipe. 2. System prompts user for recipe information. 3. User inputs recipe information. 4. Return to step 6 of normal flow. **1.2 Recommend a Recipe for Meal Plan** 1. User requests a recommendation for a recipe to choose (see 1.2.E1, 1.2.E2). 2. System gets recipe nutrition info of the recipes within the selected datetime. 3. System compares the sums of the nutrition info to the user's nutrition goals. 4. System searches the user's recipe book to find the recipes with nutrition info that if added, will bring that datetime's nutrition prediction to be more in line with the user's nutrition goals. 5. System displays a list of recipes sorted by how close each gets to fulfilling the user's nutrition goals. 6. User selects a recipe from the list. 7. Return to step 6 of normal flow.
Exceptions:	**1.2.E1 User Requests Recommendation for Past Date** 1. System informs user it can't recommend a recipe for a past date. 2. System prompts user to select a recipe eaten but not recorded on that date or to select a new date time. 3a. If User selects a recipe or inputs a new recipe they ate on that date, then return to step 6 of normal flow. 3b. Else if User selects a new date time, then return to step 4 of normal flow. **1.2.E2 User Requests Recommendation with no Recipes in Recipe Book** 1. System informs user it can't recommend a recipe with no recipes saved. 2. System prompts user to add a new recipe. 3a. If User inputs a new recipe, then return to step 6 of normal flow. 3b. Else if User cancels, system terminates use case.
Priority:	Low
Frequency of Use:	User is expected to add meals to a meal plan at a minimum on a monthly basis and up to a daily basis.
Business Rules:	BR-1, BR-2
Other Information:	1. User can cancel out of meal adding process at any time. 2. If User cancels after creating a recipe (step 3 in 1.1 or step 3a in 1.2.E2), then the recipe is still saved to the recipe book. 3. The default selected date is the current date.
Assumptions:	UC-4-AS-1: User is inputting all recipes and meals they eat into meal plan. UC-4-AS-2: User is trying to stay within their nutrition goals.

3.4.2 Stimulus/Response Sequences

1. The User indicates they want to add a meal to the meal plan.

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2. The System prompts user with meals.
3. The User selects a meal and days they plan to eat it.
4. The Systems saves meal data to system.

3.4.3 Functional Requirements

FR-4-1: When the user requests a recommended recipe and has no recipes stored, then the system shall indicate to the user that no recipe is stored and prompts the user to create a recipe.
FR-4-2: When the user indicates they want to add a meal to the meal plan the system shall prompt the user to enter which day, from the current day to any day in the future.
FR-4-3: When the user has selected a date for a meal the system shall prompt the user to select a recipe.
FR-4-4: When the user selects a recipe the system shall save the recipe to the previously specified data.

3.5 Nutrition Tracking

3.5.1 Description

The app allows users to track meal nutrition values and see personal info stats. The user can also set daily nutrition goals and edit it. (Priority: moderate).

UC ID and Name:	UC-5: Nutrition Tracking		
Created By:	David Litton	Date Created:	09/28/2024
Primary Actor:	User	Secondary Actors:	Meal Planning, USDA
Trigger:	User indicates they want to check their nutrition goal.		
Description:	User wants to see their current nutritional impact of meal in the meals in their meal plan.		
Preconditions:	UC-5-PRE-1: User has nutrition goals UC-5-PRE-2: User has been filling out the meal plan		
Postconditions:	UC-5-POST-1: User sees nutrition stats		
Normal Flow:	**1.0 Checking nutrition goal** 1. User indicates they want to check their nutrition goals. 2. System gets untracked recipes from meal plan (See 1.1) 3. System computes nutrition data of recipes using nutrition values of combined ingredients from the recipes. 4. System compiles updated nutrition goal 5. System displays new statistics of the nutrition goals to the User		
Alternative Flows:	**1.1 No new information** 1. System receives no untracked recipes from meal plan. 2. System retrieves current nutrition statistics and displays them to the User.		
Exceptions:	**1.0.E.1 No nutrition goal has been set.** 1. If the user has not specified a nutrition goal, nutrition statistics are not compared to any specific values and are simply displayed		
Priority:	Moderate		
Frequency of Use:	Approximately 2-3 usages per day.		
Business Rules:	BR-3, BR-4, B-10, BR-13		

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Other Information:	OI-1: Visuals will be specified at a later date as to how we represent nutrition tracking to the user.
Assumptions:	UC-5-AS-1. USDA has food items in database UC-5-AS-2. User has at least one recipe saved

3.5.2 Stimulus/Response Sequences

1. The User indicates they want to see nutrition.
2. The System shows User updated nutrition information.

3.5.3 Functional Requirements

FR-5-1: When the user indicates they want to check their nutrition goals, the system shall query the meal plan for food items.
FR-5-2: When the nutrition system retrieves food items from a meal plan query the system shall calculate the nutrition totals.
FR-5-3: When the user indicates they want to check their nutrition goals the system after updating the nutrition totals shall display the information graphically in a user defined manor.
FR-5-4: If the user has not added a nutrition goal and looks at nutrition statistics, then the system shall display calories by default and any other nutritional values they specify as a sum of a specified interval.

3.6 Price Comparison

3.6.1 Description

The app allows users to check the prices for items documented on their grocery list at local stores so they can compare the prices of each item between each local store to find the best price (Priority: low).

3.6.2 Stimulus/Response Sequences

1. The User indicates he wants to compare the prices of the groceries in different stores.
2. The System shows User local stores and cost of grocery list at stores.

3.7 Cooking Mode

3.7.1 Description

The app has easy timers built in that the user can use while cooking and unit conversions in cooking mode. (Priority: low).

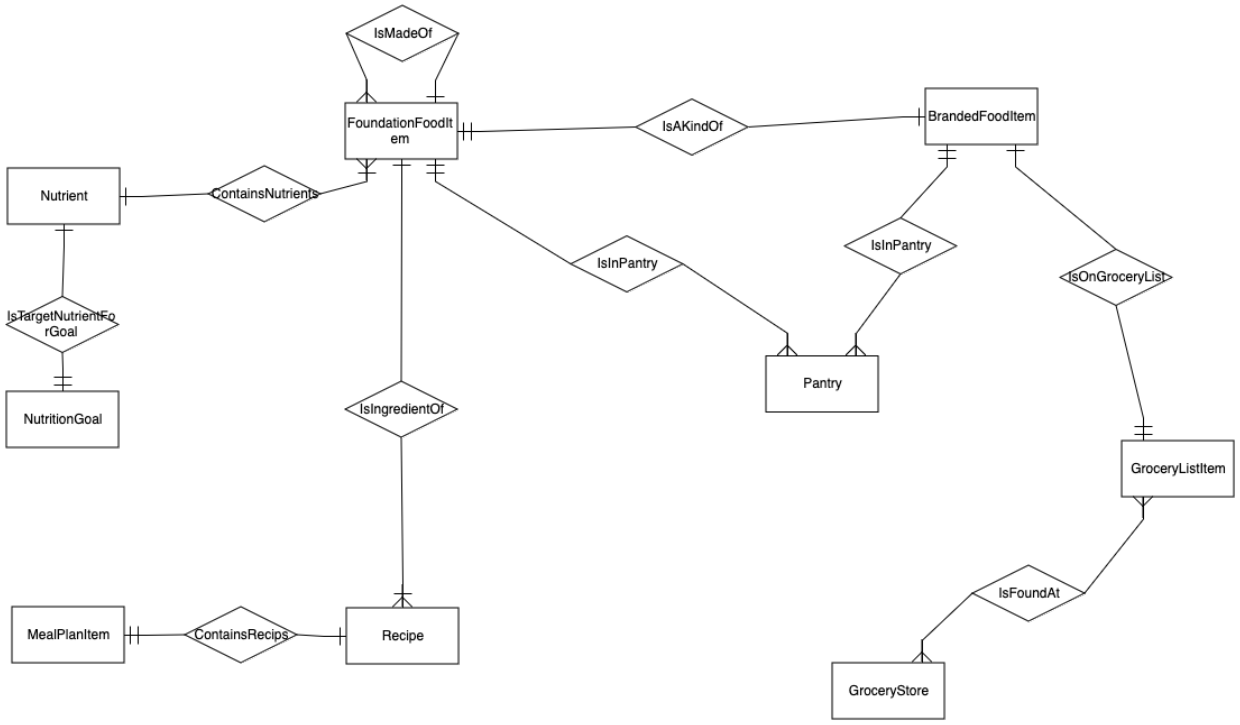
3.7.2 Stimulus/Response Sequences

1. The User indicates they want to use the cooking mode.
2. The System shows simple recipe view for cooking.

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4. Data Requirements

4.1 Logical Data Model



Logic Data Model for FillMyFridge

4.2 Data Dictionary

Data Element	Description	Composition or Data Type	Length	Values
FoundationFoodItem	General food items that are not necessarily branded and can be grouped to be a single item. It can have one or more "Input Items" which would reference other Foundation Food Items, i.e. Pasta is composed of Flour and Eggs	FDCID + Description + FoodCategory + {1:}Nutrients + FoodPortion + {1:}FoundationFoodItems		
BrandedFoodItem	Items that are packaged to be bought in a grocery store	FDCID + BrandOwner + Description + GTINUPC		

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		+ Ingredients + ServingSize + ServingSizeUnit + FoodCategory + {1:} FoodNutrients + LabelNutrients		
Recipe	A collection of ingredients, steps, and meta-data about a recipe. Nutrition and dietary restrictions will be computed rather than stored.	RecipeName + RecipeVersion + CookTime + CookingMethod + {1:}RecipeTag + {1:}RecipeIngredient + {1:}RecipeInstruction + RecipeYield		
MealPlanItem	The planned and actual times a recipe was made	Recipe + PlannedTime + ActualTime		
NutritionGoal	One single goal of many that targets a nutrient with an amount and if the goal is a target to meet and is okay to go over, or a limit to avoid exceeding	Nutrient + NutritionTarget + AtLeastTarget		
PantryItem	A singular food item being tracked with its quantity and when it was added to the pantry.	{0:1} FoundationFoodItem + {0:1} BrandedFoodItem + FoodQuantity + AddedToPantryDate		
GroceryListItem	A singular food item with the quantity to buy, the price online, the location in the store, and whether the user has added it to their cart yet	{0:1} FoundationFoodItem + {0:1} BrandedFoodItem		

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		+ FoodQuantity + Checked + GroceryLocation + GroceryPrice		
GroceryStore	A grocery store location with the user's linked account	GroceryStoreName + GroceryAddress + (GroceryAccount)		
FDCID	Identifying number for FDC	integer	10	
Description	Usually the name of the food, but can contain other information	alphabetic characters	256	
FoodCategory	General category of food	alphabetic characters	256	Vegetables, Fruits, Canned Goods, etc
Nutrient	Micro Nutrient Data	NutrientID + NutrientNumber + NutrientName + NutrientUnit		
NutrientID	USDA ID number for the given nutrient	integer	16	
NutrientNumber	The amount of the nutrient in the food item	numeric characters	16	
NutrientName	The name of the nutrient in the food item	alphabetic characters	64	
NutrientUnit	The unit for the number of the nutrient	alphabetic characters	8	
FoodPortion	String containing full text of portion size	alphanumeric characters	32	
BrandOwner	String containing the name of the owner of the brand	alphanumeric characters	64	
GTINUPC	Bar Code Value	integer	15	
Ingredients	String containing full list of ingredients as reported by brand	alphabetic characters	2048	
ServingSize	Number for serving size	integer	3	
ServingSizeUnit	Unit for Serving Size	alphabetic characters	8	
LabelNutrients	The nutrient information found on the Nutrition Label on the Branded Food Item's	Fat \+ SaturatedFat \+ TransFat		

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	packaging	\+ Cholesterol \+ Sodium \+ Carbohydrates \+ Fiber \+ Sugars \+ Protein \+ Calcium \+ Iron \+ Potassium \+ Calories		
Fat	The amount of total fat in grams	numeric characters	6	
SaturatedFat	the amount of saturated fat in grams	numeric characters	6	
TransFat	the amount of trans fat in grams	numeric characters	6	
Cholesterol	the amount of cholesterol in grams	numeric characters	6	
Sodium	the amount of sodium in grams	numeric characters	6	
Carbohydrates	the amount of carbohydrates in grams	numeric characters	6	
Fiber	the amount of fiber in grams	numeric characters	6	
Sugars	the amount of sugars in grams	numeric characters	6	
Protein	the amount of proteins in grams	numeric characters	6	
Calcium	the amount of calcium in grams	numeric characters	6	
Iron	the amount of iron in grams	numeric characters	6	
Potassium	the amount of potassium in grams	numeric characters	6	
Calories	the amount of calories in kilocalories	numeric characters	6	
RecipeName	the name of the recipe	alpha numeric characters	32	
RecipeVersion	the version of the recipe	alpha numeric characters	8	
CookTime	the estimated amount of time a recipe will take to cook	alpha numeric characters	11	12D 34H 56M
CookingMethod	A selector of the primary cooking method	alphabetic characters	32	Frying, Sautéing, Baking, Grilling,

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				Stewing
RecipeTag	A descriptor of the recipe. These will be added to by the user, but there will be initialized tags for course and cuisine	alphabetic characters	64	Appetizer, Main, Side, Dessert, Breakfast, Lunch, Dinner, Snack, Pasta, Sandwich, Salad
RecipeIngredient	The foundation or branded food item and the associated quantity.	FoundationFoodItem + BrandedFoodItem + FoodQuantity		
RecipeInstruction	The text of the instruction of the recipe	alphanumeric characters	2048	
RecipeYield	The expected number of servings	numeric characters	2	
PlannedTime	The time the recipe was planned to be eaten by the user. Formatted as YYYY-MM-DD:HH-mm	alphanumeric characters	16	
ActualTime	The time the recipe was record as eaten by the user. Formatted as YYYY-MM-DD:HH-mm	alphanumeric characters	16	
NutritionTarget	The number associated with the goal	numeric characters	4	
AtLeastTarget	If it is true, the goal is a target that needs to be met or exceeded by the user. If it is false, it is a limit not to be exceeded by the user.	boolean	1	
FoodQuantity	The amount of a food item	FoodSize + FoodUnit		
AddedToPantryDate	The date the pantry item was recorded being added to the pantry using YYYY-MM-DD format	alphanumeric characters	10	
GroceryLocation	The aisle of the selected grocery store	alphanumeric characters	16	
GroceryPrice	The price of the item at the selected grocery store	numeric characters	6	
GroceryName	The name of the grocery store	alphabetic characters	64	
GroceryAddress	The address of the grocery store	alphanumeric characters	256	

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FoodSize	The number portion of the quantity	numeric characters	3	
FoodUnit	The unit portion of the quantity	alphabetic characters	8	

4.3 Data Acquisition, Integrity, Retention, and Disposal

Acquiring data from the USDA Databases is an important aspect of our application, as all our nutrition and food data is from those databases. As these databases are updated twice a year in April and October, each time their databases are updated we will be updating our data to match the most current information available. This will ensure our nutritional information is as accurate as possible, as our users are dependent on this information in order to keep track of their nutritional information.

5. External Interface Requirements

5.1 User Interfaces

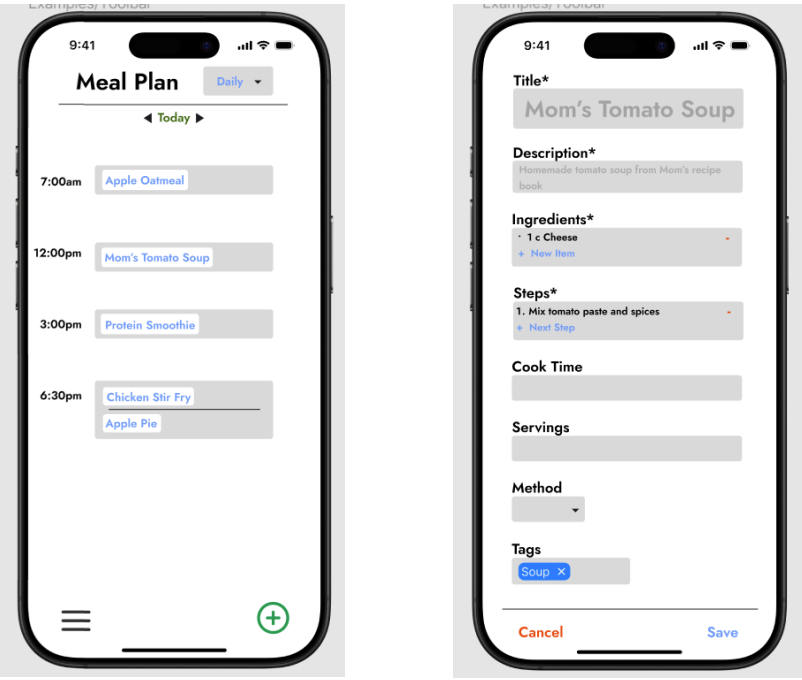


Figure 2 - Prototypes

5.2 Software Interfaces

Fill My Fridge will utilize a GET request to get recipe information from the website the user is importing a recipe from. This is done through a script we will generate that can parse through the recipe information and, based on the format of the recipe data, populate a new recipe to be stored on the users device that will be filled with the information stated in the recipe. This includes fields such

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as the title of the recipe, ingredients in the recipe and their associated amounts, the recipe's instructions, and more. Most recipe websites support these requests and follow similar formats which will allow us to populate the recipe accurately.

6. Quality Attributes

6.1 Usability

USE-1: The number of steps required to add a Recipe shall be directly proportional to the number of ingredients, instruction steps, and metadata of the Recipe itself.
USE-2: When the user indicates they have completed a form and want to save the data, the system shall add up to a 0.2 second delay to indicate to the user it is saving the data.
USE-3: The Recipe Creation Form shall be displayed as single growable paged report.
USE-4: Where a Recipe can be created, the system shall provide the same form.
USE-5: The system shall automatically use the user's live input into Ingredient fields for querying the Food Database and send the results as autocomplete recommendations.

6.2 Performance

PER-1: The system shall not take longer than 0.7 seconds to indicate a response to the user.
PER-2: The system shall have the capacity to maintain 1,000,000 records of system and user created data.
PER-3: The system shall not longer than 3.5 seconds to request and scrape recipe data from a supported external source.
PER-4: The system shall not take longer than 7 seconds to load between the user opening the app and the app being responsive to user input.
PER-5: The system shall check for data differences for premium users during the initial loading of the app.

6.3 Security

SEC-1: The system shall save any Grocery Store account data in an encrypted format while the data is at rest.
SEC-2: When the user saves Grocery Store account data to the system, then the system shall require device level authentication to access that data in the future.
SEC-3: The system shall not store any financial data to the user's device
SEC-4: The system shall salt and hash premium user passwords at rest.

6.4 Availability

AVAL-1: The premium subscriber cloud system shall be at least 90 percent available daily between 6:00 AM and 12:00 AM Eastern Time.

6.5 Integrity

INT-1: The system shall validate the user created data before saving to the device and upon retrieval for use by the system later.

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INT-2: The system shall prompt the user for verification of data changes.
INT-3: The premium subscriber system shall attempt to sync user created data to the cloud on a successful save of a new or updated user created data item.
INT-4: The system shall distinguish between draft and finished user created data.
INT-5: The system shall conform its FoundationFoodItem, BrandedFoodItem, and Nutrient Data Types as subsets of the schema defined by the USDA available on (insert swagger link here).
INT-6: The system shall save computations for items such as Calculated Nutrition as both a memoization for future calculations as well as a form of bidirectional data validation.

6.6 Robustness

ROB-1: Where there are required fields on a form, the system shall provide an example of the expected, valid data that goes into the field.
ROB-2: If the system fails while the user is filling out a form, the system shall not make an attempt to recover lost data from that incomplete form.
ROB-3: Where a form is open and the user manually closes the app, the system shall save the incomplete form as a draft.
ROB-4: While the system is responding to a user, if the system takes longer than 0.7 seconds to indicate a response to the user, then the system shall indicate to the user that it is still performing the task.
ROB-5: While the system is responding to a user, if the system has not indicated a response to the user in 5 seconds, then the system shall cancel the task and indicate to the user that there was an issue in forming the request.
ROB-6: While the system scraping recipe data from a supported external source, if the system takes longer than 3.5 seconds to indicate a response to the user, then the system shall indicate to the user that it is still performing the task.
ROB-7: While the system is scraping recipe data from a supported external source, if the system has not indicated a response to the user in 7 seconds, then the system shall cancel the task and indicate to the user that there was an issue in forming the request.
ROB-8: If the system takes longer than 7 seconds to load between the user opening the app and the app being responsive to user input, then the system shall alert the user that the app is experiencing difficulties.

6.7 Modifiability

MOD-1: A developer experienced with the system shall be able to add a new Grocery Store API client without needing to modify any of the other API clients
MOD-2: A change to the cloud provided solution shall not require changes to the client system.

6.8 Efficiency

EFF-1: The system shall require 5GB of storage space on the user's device prior to installation.
EFF-2: While the app is the active running program on the device, the system shall require 0.5GB of memory.

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7. Internationalization and Localization Requirements

LOC-1: The system shall display all text in American English across all text fields.
LOC-2: The system shall format dates in a MM-DD-YYYY format.
LOC-3: The system shall format time as a 12 hour standard formatted as hh:mm am/pm.
LOC-4: The system shall retrieve time and time zone information from the operating system.

Appendix A: Glossary

API (Application Programming Interface) - A set of rules and tools that allows software applications to communicate with each other.

Android - A mobile operating system developed by Google, used on smartphones and tablets.

Android Market - The original name for what is now the Google Play Store, where Android users can download apps.

App Store - An online store where applications for mobile devices can be downloaded, primarily referring to Apple's platform for iOS devices.

Coding Conventions - A set of guidelines and best practices for writing source code to make it more readable, maintainable, and consistent.

Context Diagram - A visual representation of a system and its interactions with external entities, focusing on data flow between the system and its environment.

Data Syncing - The process of ensuring that data is consistently updated and synchronized across multiple devices or systems.

External Systems - Any system or component outside the boundary of the main system that interacts with it, such as third-party services or APIs.

Functional - Refers to functional requirements, which define the specific behaviors, functions, or operations of a system.

GET Request – A an action that can be made to a website to ask for information from that specified website.

Grocery List - A list of items needed for purchase, often used when preparing to cook meals.

Hash – A one-way encryption scheme such that the same initial value will always have the same resulting value, but it is arbitrarily difficult to obtain the original value from the resulting value.

IOS - A mobile operating system developed by Apple, used on iPhone, iPad, and iPod Touch devices.

Kotlin - A programming language primarily used for Android development, known for its modern features and compatibility with Java.

Meal Plan - A schedule of meals to be prepared and consumed over a certain period, often for dietary or nutritional purposes.

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Mobile Device - A portable computing device such as a smartphone or tablet.

Native Development - The process of developing applications specifically for a certain platform, using its native programming languages and tools (e.g., Swift for iOS, Kotlin for Android).

Nonfunctional - Refers to nonfunctional requirements, which specify how a system performs a function rather than the specific functions it performs (e.g., performance, security, scalability).

Nutrition Goal - A specific objective related to dietary intake, such as calorie limits or nutrient targets, typically part of a health or fitness plan.

Nutritional Intake - The total amount of nutrients consumed, typically measured daily or weekly to meet dietary goals.

Premium Subscription - A paid service model where users gain access to additional features or content that are not available in the free version.

Recipe - A set of instructions and ingredients required to prepare a specific dish.

Requirement - A specification of a feature or function that a system must have to meet user needs or business goals.

Salt – A value tied to the user that is combined with their password prior to hashing to prevent users with the same passwords having the hashed values of those passwords

SRS (Software Requirements Specification) - A detailed document that describes the system's functional and nonfunctional requirements.

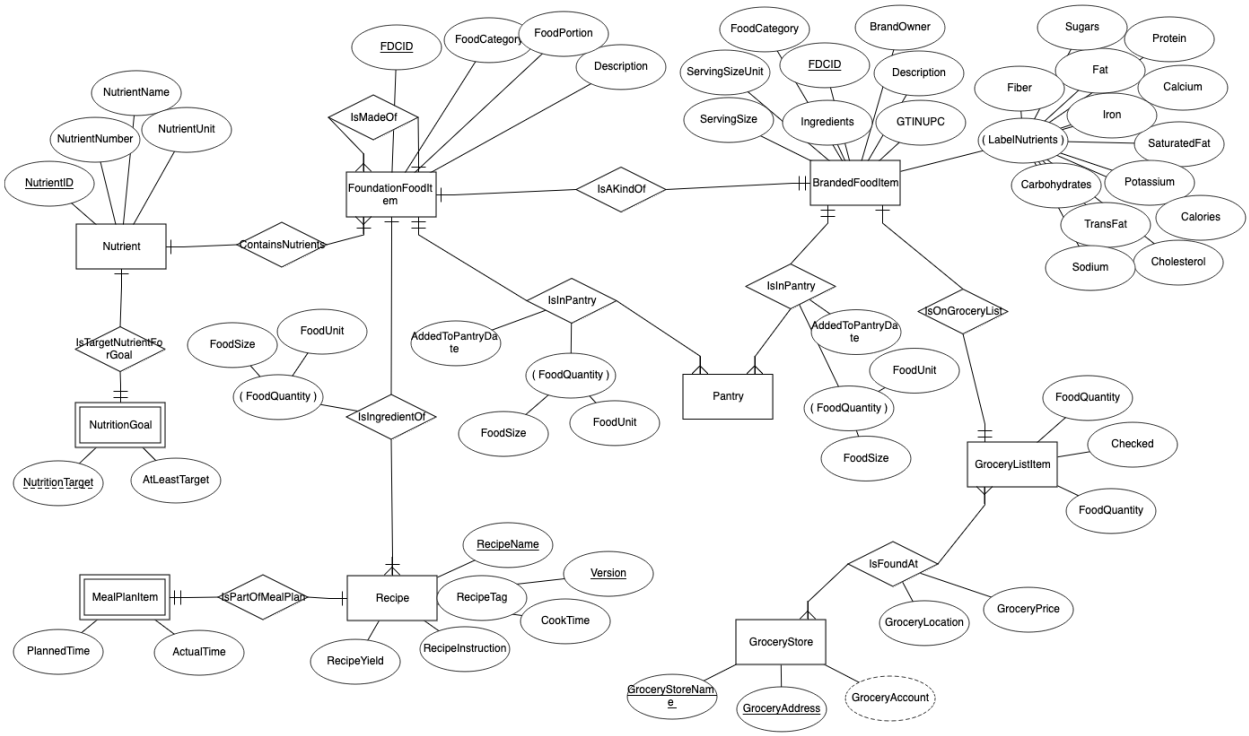
Swift - A programming language developed by Apple for iOS and macOS application development, known for its speed and safety features.

USDA Nutritional Database - A database maintained by the United States Department of Agriculture, providing detailed information on the nutritional content of a wide range of foods.

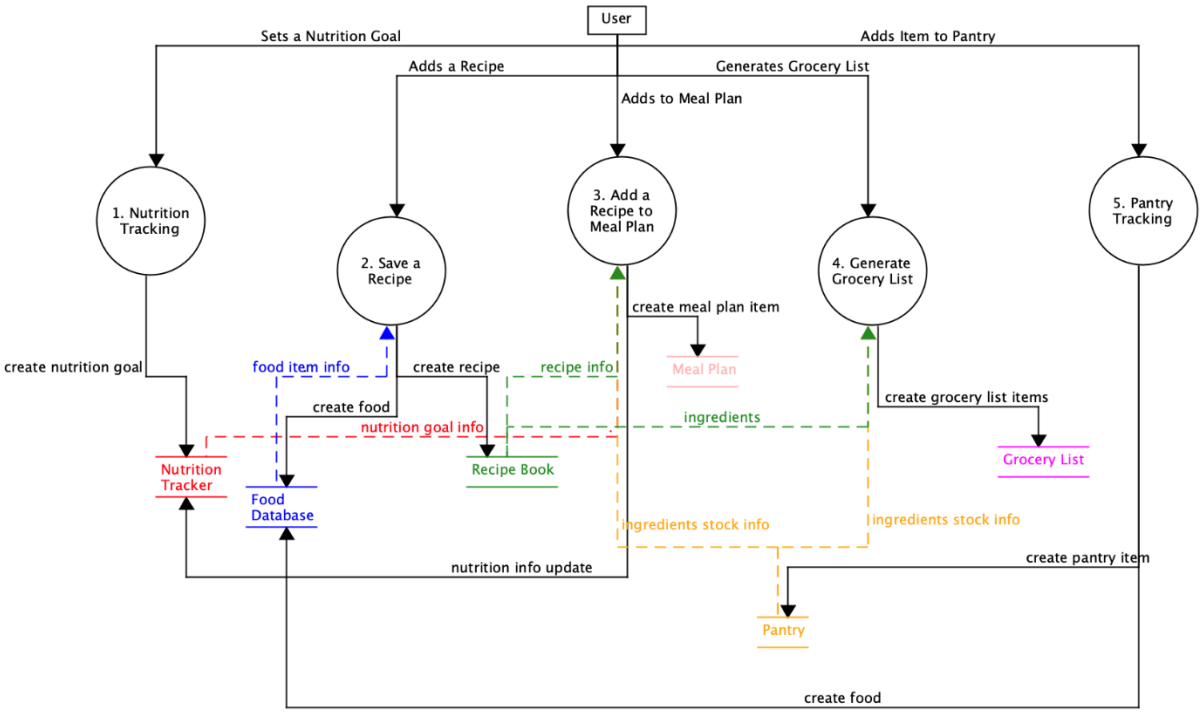
FDC – Food Data Central, the department within the USDA that handles hosting the database.

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Appendix B: Analysis Models



Entity-Relationship Diagram for FillMyFridge



Data Flow Diagram for FillMyFridge

Use Case / Entity	Recipe	MealPlanItem	NutritionGoal	PantryItem	GroceryListItem	Nutrient	FoundationFoodItem	BrandedFoodItem
Recipe Collecting	C						C, R	
Pantry Tracking				C				R
Grocery List Creation	R	R		C, R, U	C		R	R
Add a Recipe to Meal Plan	C, R	C	R	R		R	R	
Nutrition Tracking			C			C	R	

CRUD Matrix for FillMyFridge

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